

INSIDE A STAR

A STAR IS EFFECTIVELY A MACHINE FOR TRANSFERRING FANTASTIC AMOUNTS OF ENERGY FROM ITS CENTRAL CORE, WHERE THE ENERGY IS PRODUCED, OUT TOWARD ITS FIERY SURFACE. THIS JOURNEY CAN TAKE 100,000 YEARS OR MORE.

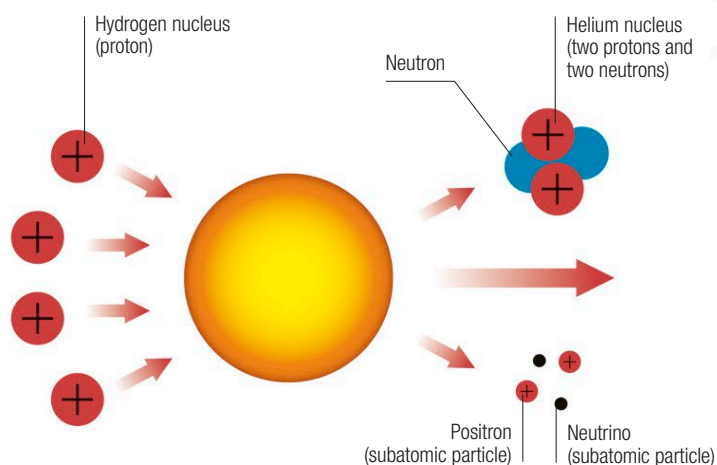
In a star, there is continuous flow of this energy from core to surface, where it escapes into space. The flow creates an outward-acting pressure, without which the star would collapse. The source of energy in the core of a star is the joining together, or fusion, of atomic nuclei (the central parts of atoms) to make larger nuclei.

Energy production and transfer

Nuclear fusion involves a tiny loss of mass, which is converted into energy. In most stars the dominant process is one in which hydrogen nuclei combine to form helium nuclei. From the core of a star, energy moves outward by radiation and convection. Radiation is the transfer of energy in the form of light, radiant heat, X-rays, and so on, all of which can be thought of as consisting of tiny packets of energy, called photons. Within a typical star, the gaseous material is so tightly packed that photons cannot travel far before they are absorbed and then reemitted in a different direction. So, energy transferred in this way travels outward in a slow, zigzag fashion. Convection carries energy toward the surface through circular motions of hot gas outward and denser cooler gas inward. Many stars contain layers, with different densities, some transferring energy by radiation, others by convection.

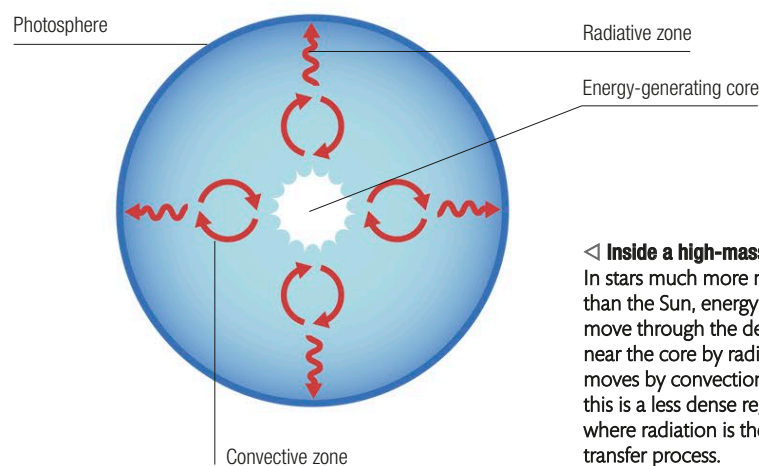
► Inside a Sun-like star

In a Sun-sized star, the core is surrounded by a radiative zone in which energy gradually zigzags outward through the emission and reabsorption of photons (packets of radiant energy). On reaching the convective zone, the energy flows to the surface by circular movements of hot gas outward, and cooler gas inward. At the star's surface, it escapes as light, heat, and other radiation.



△ Nuclear fusion in Sun-like stars

In stars about the size of the Sun or smaller, the main fusion process is called the proton-proton chain reaction. Its overall effect is to convert four protons (hydrogen nuclei) into one helium nucleus, with the release of energy and some tiny subatomic particles.



◁ Inside a high-mass star

In stars much more massive than the Sun, energy cannot move through the dense region near the core by radiation, so moves by convection. Outside this is a less dense region where radiation is the main transfer process.

Radiative zone

A region where energy slowly zigzags outward through emission and reabsorption of photons

Core

The central part of a star where energy is produced by nuclear fusion reactions

